Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Agentic AI for Business Process Automation

TGS Ref No: TGS-2024045801

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	31 July 2026	Initial release — 2-day (14-hour) lesson plan for Agentic AI for Business Process Automation, aligned to the 5 hands-on labs and TSC AER-TEM-3026-1.1.	Dr. Alfred Ang
1.1	1 August 2026	Day 2 rescheduled so the final assessment follows instruction within the training day; daily timing now driven from a single source shared with the slide deck; assessment time stated separately from instructional time.	Dr. Alfred Ang
1.2	1 August 2026	Added a screenshot of each lab's application — the four lab websites with their chat widgets open, plus the Lab 1 n8n canvas and sample flyer — to the slide deck and the Learner Guide.	Dr. Alfred Ang
2.0	8 August 2026	Slide deck restyled to the house reference format (visual concept cards, chevron flow strips, comparison panels and takeaway bands). Lab step slides removed from the deck; the click-by-click steps now live only in the Learner Guide, and each lab carries a rendered screenshot of its own n8n canvas. Concept coverage expanded: trigger nodes, data-processing nodes, the four components of an AI agent, instruction design, model parameters, MCP, webhooks (GET/POST, Test vs Production URL) and human-in-the-loop with guardrails.	Dr. Alfred Ang

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Course Information

Course Title	Agentic AI for Business Process Automation
WSQ Course Reference	TGS-2024045801
TSC Title / Code	Artificial Intelligence Application · AER-TEM-3026-1.1
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	Day 1: 9:30 am – 6:30 pm. Day 2: 9:30 am – 6:30 pm instruction, assessment to 8:45 pm. Each day is 8 instructional hours: a 1-hour lunch is excluded, and the two 15-minute tea breaks are counted within the instructional time.
Mode	Instructor-led, hands-on workflow-automation labs built in n8n
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain what n8n and AI agents are, and build automated workflows using triggers and actions.
- LO2: Implement AI agent applications that are triggered by webhooks and integrated with business systems.
- LO3: Design agentic chatbot workflows that operate safely within business and regulatory constraints.
- LO4: Apply human-in-the-loop approval so that sensitive actions are authorised by a person.
- LO5: Deploy Retrieval-Augmented Generation (RAG) to ground an agent in an organisation's own documents.

Skills Framework Alignment

This course is mapped to the TSC Artificial Intelligence Application (AER-TEM-3026-1.1).

Code	Ability / Knowledge statement
A1	Implement AI applications in collaboration with technology service providers
A2	Deploy AI workflows according to plan
A3	Identify and report any issues with the AI applications and data collected
K1	Background and basics of AI
K2	Fundamental concepts and methods of statistics

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	and programming for data science
K3	Mathematics and computing theories
K4	Machine learning techniques and applications
K5	Principles of data management
K6	Software development methodologies
K7	Types of AI applications
K8	AI implementation procedures

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on n8n build tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The Written Assessment covers the Knowledge (K) statements; the Practical Performance covers the Ability (A) statements.
- Final assessment is conducted at the end of Day 2.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — n8n, AI Agents and Agentic Chatbots

Time	Duration	Topic / Activity (with slide references)
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, course outline, learning outcomes, Skills Framework alignment and mandatory digital attendance (AM). Slides 2–19
10:00–11:00	60 min	**Foundation — Why agentic AI: the evolution of AI, agentic vs generative AI, from RPA to Agentic Process Automation, business value and use cases. Slides 20–26**
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	**Topic 1 — Overview of n8n and AI Agents: what n8n is, hosting options, credentials, node types, trigger nodes, data-processing nodes, JSON and expressions, execution modes. Slides 27–39**
12:30–13:30	60 min	Lunch break
13:30–14:45	75 min	**Topic 1 (continued) — AI agents: the four components (model, memory, tools, instructions), popular LLM providers, model parameters and temperature, instruction**

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		design with worked examples, and MCP. Slides 40–49
14:45–15:45	60 min	**Topic 1 (continued) — Webhooks: the inbound URL, webhook vs HTTP Request, GET vs POST, Test vs Production URL, webhook configuration and structured output. Slides 50–57**
15:45–16:00	15 min	Tea break
16:00–17:00	60 min	Hands-on Lab 1 — n8n Trigger and Action (Event Flyer with QR Code). Slides 58–61
17:00–18:15	75 min	Hands-on Lab 2 — Webhook and AI Agent (Marina Trust Bank Onboarding). Slides 62–64
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Instructional time: 480 minutes (8 hours). The day runs 9:30 am - 6:30 pm: the 1-hour lunch is excluded from instructional time and the tea breaks are counted within it.

Day 2 — Human in the Loop, RAG and Assessment

Time	Duration	Topic / Activity (with slide references)
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	**Topic 2 — Agentic Chatbot and Human in the Loop: human in, on and out of the loop; how an approval suspends a running workflow; when a human belongs in the workflow; and guardrails. Slides 66–70**
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	Hands-on Lab 3 — Chatbot Webhook (Investment Advisor): the contact gate and the four compliance probes. Slides 71–73
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on Lab 4 — Human in the Loop (Client Rapport Assistant): the approve and decline paths. Slides 74–76
15:00–15:15	15 min	Tea break
15:15–16:30	75 min	**Topic 3 — Retrieval-Augmented Generation: why retrieval rather than a bigger prompt, the RAG pipeline, tokenization and embeddings, chunking, vector stores and grounding. Slides 78–86**
16:30–18:00	90 min	Hands-on Lab 5 — RAG

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		Customer Care Chatbot (Cook & Bake Academy): ingestion and the grounded chatbot. Slides 87–90
18:00–18:30	30 min	Course recap, taking it back to work, Q&A. Slides 91–93
18:30–18:45	15 min	**Briefing for Assessment and Assessment digital attendance**
18:45–19:45	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
19:45–20:45	60 min	**Practical Performance (PP) — hands-on n8n build tasks, 1 hour, open book. TRAQOM survey and Assessment digital attendance**

Instructional time: 480 minutes (8 hours). Assessment: 2 hours (WA 1 hour + PP 1 hour), plus a 15-minute briefing. The day runs 9:30 am - 6:30 pm: the 1-hour lunch is excluded from instructional time and the tea breaks are counted within it.

Total instructional time across the course: 16 hours (2 days × 8 hours), plus 2 hours of assessment (WA 1 hour + PP 1 hour) and a 15-minute assessment briefing.

Lab Reference (aligned to topics and learning outcomes)

Topic	Outcomes	Labs	Deck slides
Topic 01: Overview of n8n and AI Agents	LO1 · LO2	Lab 1, Lab 2	Slides 27–65
Topic 02: Agentic Chatbot and Human in the Loop	LO3 · LO4	Lab 3, Lab 4	Slides 66–77
Topic 03: Retrieval-Augmented Generation (RAG)	LO5	Lab 5	Slides 78–90

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